



ELEC S411F (2025/26)

Electronic and Computer Engineering Project

Interim Report

FYP Matching System

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1. ABSTRACT

In this project, we describe the design and development of the FYP Matching System for Electronic and Computer Engineering (ECE) and Computer and Cyber Security (CCS) students of Hong Kong Metropolitan University. Currently, the allocation process depends on extensive coordination among multiple stakeholders, using dispersed communication methods and a 'black box' matching mechanism. In response to the identified challenge, this initiative puts forward a web application built upon the MERN stack, encompassing MongoDB, Express.js, React, and Node.js. This application incorporates three distinct portals, each tailored for specific user groups: students, supervisors, and administrators. The system employs a preference-driven matching algorithm, which considers students' prioritized project selections and leverages GPA as a tie-breaker mechanism. Among its principal functionalities are real-time tracking of project popularity, a drag-and-drop interface for preference ranking, automated management of submission deadlines, and a manual resolution procedure for students who remain unmatched. The project adheres to a three-layer client-server architectural model, fortified by role-based authentication and protected routing protocols. Currently, the system's operational modules include user authentication and login processes, a student portal featuring the drag-and-drop ranking functionality, the core matching algorithm, and fundamental administrative controls. For subsequent development phases, plans involve finalizing the supervisor and administrator portals, enhancing data visualization capabilities, establishing a comprehensive notification system, and conducting both usability and load testing.

2. INTRODUCTION

Final Year Project (FYP) plays an important role in the ECE and CCS of Hong Kong Metropolitan University. It is a two-term, 10-credit module that allows students to develop the skills necessary to undertake an independent research or development project. Regardless of the university or major, the student is required to complete his/her FYP so as to round off the knowledge obtained from the major. An important and sometimes complex part of the exercise is the assignment of projects to supervisors and students. Prior to undertaking their projects, the students must select an appropriate topic of the major for their FYP, in a proposal, matching, and clearing system. The process, while functioning, relies mainly on manual coordination and dispersed communication like email and forms without having a single platform for decision making.

These drawbacks of using the traditional manual method of assigning final year projects involved:

1. **Lack of Transparency:** The students cannot know in advance how well-received some projects may be or whether any of the professors offering projects are in high demand. The students usually have no prior knowledge about their respective final year projects or the professors offering them. Thus, they are left with no option but to make an informed decision based only on the description of the project.
2. **Inefficiency for Administrators:** It is natural that dealing with student preferences, matching according to GPA and their own interest, and handling the clearing process for those who are not matched takes up much time and effort and involves human error. Even if certain components of matching are mechanized under the existing system, administrators are still faced with the inconvenience of handling special cases, exceptions, and the overall coordination, which continues to be a burdensome administrative task.
3. **Limited Visibility for Supervisors:** The "big picture" is not easily visible to teachers. They lack insight into how their projects are relatively more or less popular than

others, which would provide valuable feedback to influence subsequent project design. Another significantly reported issue is the "silo effect"; when supervisors submit official and present their project titles, they lack a systematic way of seeing what other faculty members are proposing. This have the tendency to lead to imbalance project difficulty and scope, and in some instances even replication or duplication of project ideas, ultimately reducing the diversity and balance of available alternatives for students.

4. **Delayed and Opaque Communication:** Students and teachers may not be aware of their assignment status or the overall progress of the allocation process until the final announcement. The students and faculty members remain in the dark concerning the status of the assignments and progress being made until the very last moment when they receive an announcement. This creates unnecessary uncertainty that does not allow them to plan anything or prepare themselves for possible issues.

2.1 AIMS AND OBJECTIVES

The main objective of this research is to create, design, and implement a web-based FYP Matching System. The web-based FYP Matching System will act as a digitalized platform to help automate the entire process of allocating FYPs to students. This system will seek to address the challenges associated with the existing fragmented and inefficient system currently in place. It will be an advanced FYP Matching System that improves the efficiency of the matching period.

2.1.1 Revised Objectives Based on Implementation Progress

While the primary objective of this research project continues to be the development of an FYP Matching System via a website platform, the current progress made within this project enables us to re-evaluate the original objectives and present them with more precise indicators.

In view of this objective, the research project will proceed through the following objectives:

1. To Develop a Centralized and Role-Based Web Application

To develop a comprehensive and cohesive web portal for the use of all parties concerned. The software will incorporate three separate user interfaces, which would include those for Students, Supervisors, and Administrators, and would make it easy for users to access the information required by them.

2. To Implement a Robust and Transparent Matching Algorithm

To develop an algorithm for the matching of the cores such that the allocation of projects to students becomes automatic. The system will primarily focus on how students prioritize their choice of projects. In cases where two students desire the same project, then GPA shall become the decisive factor.

3. To Enhance Decision-Making with Real-Time Data Analytics

In order to incorporate the data visualization capabilities that provide insights in real time. The students will be able to determine how popular each project is (in terms of number of project choices made) and thus select their projects based on such data. Supervisors will get an overall picture of all the projects available for selection.

4. To Streamline Administrative Workflow and Management

To create a robust administrative system that will take care of some important functions automatically. These include handling of user accounts, project due dates, initiating the matchmaking process, monitoring of events, and taking care of students who have not been matched yet.

5. To Ensure System Reliability and Usability

To make sure that the final product is not just efficient but also strong and friendly enough to use. This means rigorous testing to ensure system stability during maximum loads (for example, during the deadline time) and creating a user-friendly interface [9], which needs very little training from all users.

Table 1 summarizes the current status of each original objective:

Objective	Original Status	Current Status	Remarks
Develop a Centralized and	Planned	Implemented	A role-based web portal has been

Role-Based Web Application			implemented using React 18 and Vite, with distinct interfaces for Students, Supervisors, and Administrators.
Implement a Robust and Transparent Matching Algorithm	Planned	Implemented	An enhanced Gale–Shapley-based matching algorithm has been developed, incorporating GPA-based tie-breaking and project capacity constraints for fair allocation.
Enhance Decision-Making with Real-Time Data Analytics	Planned	Partially Implemented	Real-time project popularity counters have been implemented in the Student portal, while full statistical charts and dashboards remain under development.
Streamline Administrative Workflow and Management	Planned	Partially Implemented	Core administrative features such as project review and manual allocation are functional; deadline management and notification

			mechanisms are still pending.
Ensure System Reliability and Usability	Planned	Partially Implemented	Basic functional testing has been carried out; comprehensive load testing, security testing, and user acceptance testing have yet to be completed.

Table 1

2.2 ORGANIZATION OF THE REPORT

The structure of the paper comprises a succession of important chapters dedicated to the description of the background, development, and implementation plan of the suggested FYP Matching System in a step-by-step manner. The first section of the paper (Introduction) defines the background, motivation, general goals, and objectives of the project. Afterward, Section 2 (Literature Review) provides an evaluation of relevant systems and a discussion about the need for the selection of the suggested technologies (MongoDB, Node.js, Express.js, and React) in order to build a more efficient platform for the purpose of the project. The methodology used for building the suggested system is presented in Section 3 (Methodology), including the system design, user interfaces, and workflow. The fourth section (Experiment) addresses the prototype development. The fifth section (Results and Discussion) presents the milestones achieved so far, as well as a discussion about the current situation. Finally, the conclusion of the work and plans for further research and improvements are outlined in Section 6 (Conclusion and Further Work).

3. LITERATURE REVIEW

A good web application in the current world would need an appropriate choice of technology when it comes to designing the database, backend, and front-end. After choosing from several others, the MERN Stack (MongoDB, Express.js, React, Node.js) [3, 6, 12] was chosen due to its ability to meet the needs of flexibility, efficiency, and performance in developing the system.

3.1 TECHNOLOGY STACK SELECTION AND JUSTIFICATION

3.1.1 MongoDB: The NoSQL Database

MongoDB is an open-source document database designed to create high-performing web applications. [7] With the flexible approach of its schema design, it is preferred by software development teams adopting agile approaches. Supporting various programming languages' drivers, it enables users to begin building their application right away without setting up a database first. Why do we choose MongoDB instead of other databases systems? This document-based database suits perfectly for the FYP system's data storage [13]. The information about the Project may be saved in a Project document. It may contain all the required information like title, description, and supervisor. The information about Student project preferences may be stored in a Student document. The flexible nature of the schema makes it convenient to modify the schema on the fly. Also, JSON format suits perfectly for the JavaScript-based backend and frontend architecture.

3.1.2 Node.js & Express.js: The Backend Foundation

Node.js is an open source runtime environment that runs the JavaScript code on servers outside the browser environment. [8] Traditionally, JavaScript can only be used in the client side web development in browser environment, but through Node.js JavaScript can be used in the server-side and networking. Express.js, also known as express, is the most widely used framework among all other frameworks available for developing web applications and APIs. [2] It is considered to be the de facto standard server framework for node.js. It is also a minimal web application feature framework that provides an abstracted version of basic web applications without making any node.js features obscure. Node.js and Express.js over alternatives such as Django and Spring boot framework primarily because using Node.js we can use JavaScript

language for both client side and server-side web development. In simple words, we can use JavaScript for everything, and not having to shift from one language to another during development saves much hassle. Plus, it efficiently deals with numerous connections, which comes in handy at times like preference submission [1].

3.1.3 React: The Frontend Library

React, also referred to as React.js or ReactJS, is a free, open-source front-end JavaScript library that seeks to enable seamless creation of user interfaces using components [10]. It is maintained by Meta (previously Facebook) along with the help of individuals and organizations. React has been preferred over others such as Vue.js and Angular due to the capability to create interfaces for students, teachers, and admins that are consistent but differentiable from each other. [14] Additionally, React's Virtual DOM ensures fast performance and smooth transitions, which is crucial when displaying dynamic content like real-time popularity counts. Lastly, React is more intuitive than Angular, which is a full-fledged framework.

3.2 ANALYSIS OF EXISTING AND ALTERNATIVE ALLOCATION SYSTEMS

This segment will examine the existing HKMU system and compare it to some other types of automated allocation systems. Having examined these choices, one will develop a clearer image as to why the system that I developed was designed that way. The idea is to retain the best elements of the present approach, address weaknesses, and incorporate useful components from other existing approaches that have proved effective.

3.2.1 The Current HKMU System: A Batch-Processing "Black Box"

The current allocation system in place at HKMU is officially broken down into three clear stages, according to the student handbook. However, when viewed in terms of technology and usability, each stage lacks integration and clarity.

1. **Proposal Phase:** In this first phase, the supervisory staff gives the names for the research projects. But they do not have a common platform where they can check what the rest of the faculty is working on. This may cause some of the projects to overlap.

2. **Matching Phase:** Here comes the automation, although not without some mystery as well. Students examine the list of projects and prioritize their options by filling out a straightforward form, such as Google Forms. After that, once the deadline passes, the magic happens through some undisclosed matching algorithm, considering both student preferences and GPA, yet without disclosing any details to anyone or updating participants about the process.
3. **Clearing Phase:** If no match is found for some students on any of the projects they prefer during the Matching Phase, then a clearing phase takes place. However, this phase cannot be completed without assistance from the administrator in allocating students to the openings. It is difficult to understand the process of this stage, and the students are not privy to it.

Advantage: The three phases of the system make the administrator end simple to understand. Furthermore, the automation of the Matching Phase reduces complexity compared to performing all tasks manually.

Disadvantage: The major limitation of this system is its lack of clarity and coherence. The three modules "Proposals", "Matching", and "Clearing" operate independently from one another. Since users have to employ various platforms for each module, it becomes difficult to make future predictions. This, in turn, causes fear and uncertainty about the system's credibility among its users.

3.2.2 Considered and Rejected: The First-Come-First-Served (FCFS) Model

Automating an FCFS model to improve it, but I chose not to do that. FCFS is fast at providing feedback and straightforward to implement. However, it is not entirely fair since it emphasizes connectivity speed more than skills and suitability of students. As the approach was already fair enough to use in educational settings, this model became a good choice for implementation.

3.2.3 Considered and Adapted: Principles from Stable Matching Algorithms

Applying standard matching algorithms as provided in Computer Science literature, such as the Gale-Shapley Algorithm [4] created by Gale and Shapley, finds a matching among two different sets by proposing matches in one direction using the preference list of each member. However, the approach is sophisticated and hard to comprehend for most. Furthermore, all participants have to submit their full preference list which cannot be achieved in our case.

Thus, my algorithm does not rely on an elaborate matching approach. Rather, it utilizes the main criterion "respecting ranked choices" from Stable Matching Theory. The algorithm employs a simple, rule-based approach based on ranking and GPA tie breaker criteria [11]. Therefore, it is fair and comprehensible for all participants, offering a balance between accuracy and simplicity.

3.2.4 Comparative Analysis and Positioning of the Proposed Solution

Having analyzed the current process as well as alternatives, specific design requirements are identified for the proposed FYP Matching System. While the batch processing performance must be maintained similar to the current HKMU system following the deadline period, usability and comprehensibility of the system must also be significantly enhanced as it is currently quite difficult to comprehend and relate.

Aspect	Current HKMU System	Proposed FYP Matching System
Transparency	"Black-box" model. Undisclosed matching logic.	"Glass-box" model. Clear, rules-based algorithm (Preferences + GPA).
Supervisor Insight	Siloed view. No visibility into other projects.	Holistic dashboard. Ability to view and compare all projects.
Student Decision-Making	Blind submission. No data on popularity or competition.	Informed strategy. Real-time popularity statistics for projects.
User Experience	Fragmented workflow. Relies on separate, generic tools.	Unified platform. A single, integrated web application for all tasks.

Communication	Prolonged uncertainty. Status unknown until final release.	Process visibility. Clear indications of the current phase and status.
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Table 2

In conclusion, the proposed FYP Matching System is designed to address the issues in the existing HKMU matching system. This novel approach incorporates the principles of transparency, user empowerment, and user-centeredness, while retaining the fairness and batch processing aspects of the existing system and circumventing the flaws of simple approaches such as FCFS and pure stable matching systems.

4. METHODOLOGY

The current part will focus on both theoretical and practical aspects associated with the system itself, basic requirements for the system that had been analyzed earlier in the corresponding chapter, as well as interfaces of the system designed in this project. The HKMU FYP Matching System proposed by the author has a role-based design interface comprising three different user interfaces according to the specific three-tier architecture [5].

4.1 SYSTEM ARCHITECTURE AND DESIGN

This section will deal with the overall architecture of the system, comprising three-tier system architecture, interactions between components, security issues, as well as database design within the FYP Matching System.

4.1.1 Three-Tier Architecture

The system makes use of the well-known three-tier client/server architecture [5] to organize the application in three layers – Presentation Layer, Business Logic Layer, and Data Layer. Such architecture promotes modularity, scalability, and easy maintenance during both the development and deployment phases.

The Presentation Layer takes care of all the operations related to the user interface and operates solely on the client's side (web browser). The Business Logic Layer handles all the processing of requests, execution of business logic, and communication between other layers. Finally, the Data Layer deals with the storage and management of database content through MongoDB.

4.1.2 Presentation Layer (Client)

For the creation of the frontend of our system, we decided to choose React framework, along with using Vite as a build system. Our presentation layer has a number of architectural elements that guarantee the responsiveness of UI.

Component-based architecture enables the decomposition of the user interface into the reusable modules within three different directories in the portal: Student Directory consists of modules for proposing ideas, browsing through projects, managing preferences and viewing results. Teacher Directory includes modules for reviewing students' proposals, managing the projects proposed by teachers, and viewing supervision assignments. Admin Directory consists of modules related to system configurations, control of matches, and managing user accounts.

React Router is used to implement a role-based routing solution that ensures redirecting users to appropriate pages depending on their role in the system. Protected routes are used to secure pages with sensitive information, and access to protected pages is restricted depending on whether the user belongs to certain roles.

There is a mandatory change process for password during authentication in this system. The user is supposed to change his password immediately after signing up in the system and use the newly created password while logging into the system. The user is not supposed to use his first password at any time, and the shortest password allowed is 8 symbols long.

4.1.3 Application Layer (Server)

Node.js and Express.js framework are utilized to develop the server application layer, thus providing an efficient development environment. There are many services used in combination within the application layer for the purposes of processing users' requests and performing business logic operations.

This part presents some information about important endpoints in the server. Authentication endpoints are responsible for the processing of login requests and password changing. Students' endpoints take care of project browsing, preference sharing, proposals, and assignment verification. Teachers' endpoints include functions like adding new projects, evaluating proposals, and supervising. Admin endpoints include configuration, deadline setting, matching, and account batches management.

Our matching engine will be central to the server code, as it will involve the utilization of our algorithm in order to perform the process of matching students and projects, based on their

preferences. Matching will take into consideration the rank of preferences, GPA score, and availability.

4.1.4 Data Layer (Database)

The reason why MongoDB is used is because of its flexibility and ability to integrate with the JavaScript environment easily. The Mongoose Object Data Modeling (ODM) library provides schema-based data modeling and validation for MongoDB [16].

Four main collections are included in the database. To start with, the Users collection has data related to all users registered on the application. Here, a field named role, which could be either Student, Teacher or Admin. As for Students' collection, apart from the usual credentials fields, some other important fields include GPA, Major, (either ECE or CCS) and preferences. For teachers, fields such as department, majors taught are some of those included in the collection. Lastly, the Projects collection includes all the registered projects. These projects are associated with an extra field that identifies whether it has been created by a teacher or a student.

4.1.5 Security Architecture

Different security mechanisms are used in this project to protect user data and grant access accordingly.

Passwords are hashed using bcryptjs. Passwords are always hashed before storing and there are password length limitations in this application. Additionally, the user is prompted to change their password after their first login by changing the value of the mustChangePassword property. The user cannot use the same password that was used during registration.

The access of functions depends on user roles. This ensures that every user can access functions relevant to their positions. Students can view the projects available, select their preferences, and check the allocation status of their tasks. Teachers can edit their projects and evaluate the proposals from the assigned students. Administrative users have unlimited access to the system's configuration and management.

The requests made to API endpoints are verified to determine whether the inputs are valid.

4.2 DATABASE DESIGN

4.2.1 Database Schema Design

There are four main collections in the MongoDB database, where all data of the system is stored. The following part presents the design schema for each of the collections.

4.2.2 User Collection

The Users collection consists of important information regarding all users of the system, where they are classified based on the role they play as being either a student, teacher, or admin.

All user profiles have an email attribute used as a unique identifier, a password attribute that stores the bcrypt hashed password, and role attribute that shows either student, teacher, or admin. For student users, there are additional fields such as studentId for the official student number, gpa for tracking performance, and major for determining whether the student is pursuing a degree in either Electronics and Computer Engineering or Computer and Cyber Security. preferencesSubmitted is a boolean to determine whether the user has committed to their choices for the projects. proposalSubmitted is a boolean to indicate whether the student has submitted their proposal. assignedProject references the matched project after executing the matching algorithm. proposedProject refers to the self-proposed project by the user. proposalStatus indicates the status of the user's submission of the project proposal, and teacherNotes is an array containing notes written by the reviewing teachers regarding the student's proposal. mustChangePassword determines whether the user should change their password after logging in. initialPassword contains the initially generated password of the user.

Teacher users have the following attributes: teacherId, which is the unique identifier, department, which specifies the academic department, and major, which is the specialization.

4.2.3 Project Collection

The Project collection holds information about all of the FYP projects, whether teacher-proposed or student-proposed.

The main fields include code for the unique identifier of the project, title for the project name, and description for detailed information about the project. The type field distinguishes between a teacher-proposed project or student-proposed project. Capacity refers to the maximum number of students that can take the project, and it defaults to one student. Skills holds the array of technical skills required for the project. Category and department categorize the project.

Status tracking fields include status for the project's overall status, proposalStatus for specific student proposals, and isActive for visibility control purposes.

In case of teacher-proposed projects, supervisor is a reference to the user table that holds data for the teacher, supervisorEmail is the email address used for displaying purposes, and teacherReviews is the array of review records showing who has reviewed the project.

Popularity tracking fields include popularity, which is the count of students who have added the project to their favorites list, and assignedCount, the count of students who have been assigned the project.

4.2.4 SystemSettings Collection

SystemSettings is a singleton document that contains global system configuration information.

deadlines stores timestamp variables for studentSelfProposal, preferenceSubmission, teacherProposalReview, and teacherSelfProposal deadlines, which define what functions will be available during specific periods of time. matchingCompleted stores a Boolean value of true or false, depending on whether the process of matching was completed during the current cycle. CurrentPhase is a string variable storing an active system phase.

4.2.5 Database Relationships

The referential integrity in the collections is achieved using ObjectId references. The User collection references the Project collection as supervisors. The User collection references the Project collection as proposedBy students. The User collection references the Project collection

as assigned supervisors following the matching process. The Project collection references the User collection as assigned projects.

4.3 ADDITIONAL USER INTERFACE FEATURES

In addition to those mentioned above, some other interface elements and interactions that improve user experience can be used in all portals.

4.3.1 Project Filter Interface (Student)

The project search page gives several filters to find suitable projects for students. Firstly, there is a search bar where students can enter key phrases to find their desired projects in project titles, project descriptions, and project supervisors' names. Secondly, the skill filter offers a multi-select box where users may choose only projects corresponding to these skills. The supervisor filter is a dropdown list containing all supervisors which can be used for searching. The status filter gives two states — only show active projects or show all projects. Sorting criteria are set to sort by popularity (number of students who chose this project), alphabetical order by title, and sorting by supervisors' names.

Every project card contains a project title, supervisor name, description snippet, skill tags, and number of likes.

4.3.2 Preference Management Interface (Student)

The My Preferences tab shows the current preferences in the student's list, with clear visual cues regarding their order. The preferences themselves contain information on the title of the project, its supervisor, and rank number. There are drag handles present on the left side of each preference, indicating that items in the list can be reordered. Alternatively, students have access to Up/Down buttons present on the right side. Clear All button allows clearing the preference list, while the Submit Preferences button confirms selected preferences and opens a confirmation window.

Once preferences are submitted, the window displays confirmation of the submission and switches to a read-only view of submitted preferences, disabling the Submit button.

4.3.3 Results Interface (Student)

The Results page gives students clear indications regarding the outcome of their assignments. If no results are received from the matching process yet, the student will receive an indication informing them that results will be shown once ready. Once the results are calculated, a loading animation will indicate the process, which is followed by displaying all the data regarding the assigned project.

4.3.4 Project Creation Interface (Teacher)

For teachers creating new projects, there is a form with fields including title, description, skills selection, major, and capacity. Skills selection enables the entry of more than one skill selected from an existing list. Major sets the criteria of students who can view the proposed project. Capacity is preset at one and can be adjusted if the project needs more than one student.

List of Existing Projects displays all projects created by the particular teacher along with editing and deleting options.

4.3.5 Proposal Review Interface (Teacher)

The proposal review interface lists all the student proposals according to the teacher's major expertise. For each proposal listed, the system displays the student name, student ID number, major, grade point average (GPA), title of the proposal, and time of submission. Teachers are expected to review the proposals and decide whether to approve or disapprove the proposals. Approved proposals turn into projects managed by the particular teacher.

4.3.6 Account Creation Interface (Admin)

For the addition of accounts, there is an admin account creation system that operates in two different modes. The administrator will be able to switch between Student Account Creation Mode and Teacher Account Creation Mode using tabs.

Under the Single Creation Mode, the administrator will fill up a form with all the information required. Under Batch Creation Mode, administrators will be able to click a template download button to download a CSV template file. Once the administrator has filled up all user entries on the file, he/she can then upload the file using the file input field. There will be a data preview before submission and a report on all successes and errors made.

For both student and teacher accounts, the email used will automatically generate based on the institutional format of email generation. For students, it will be s (the first letter of student), followed by student id and then @hkmu.edu.hk. Similarly, for teachers, it will be t (the first letter of teacher), followed by teacher id and then @hkmu.edu.hk.

4.3.7 Matching Control Interface (Admin)

The admin matching control panel is used for managing the matching process centrally. A Start Matching button starts the matching process. There is a Reset Server button that resets all preferences and assignments in order to give users the chance to do the process anew. Live statistics show the current number of total students, submitted preferences, matched students, and unmatched students. Statistics are categorized by majors (ECE and CCS).

There is a refresh button that lets administrators refresh statistics without running the matching algorithm.

4.3.8 Final Assignment Interface (Admin)

The final assignment interface is used for handling the clearing phase, where some students will be assigned manually because of their unmatched status. There is a table for unmatched students with pagination support when there are many unmatched students. There is also a table for matched students with current assignments and an available projects panel with projects that have open slots with lazy loading support.

Administrators can assign manually a selected unmatched student and a selected project by clicking on the Assign button. Administrators can also edit manually a selected matched student by assigning them to another project. There is also a Clear Assignment button to clear current assignments.

4.3.9 Batch Assignment Interface (Admin)

Where many students need to be assigned manually, the interface is designed to enable batch processing. It is possible for an administrator to select more than one student who has not been matched and employ the auto-assignment facility to automatically assign each selected student to a project that still has vacant slots based on their area of specialization.

4.3.10 Data Export Interface (Admin)

This part allows users to download various reports from the system. The export capabilities allow the user to download matching results in CSV format, a list of all students in CSV format, a list of all projects in CSV format, and a list of all teachers in CSV format.

4.4 SYSTEM INTERFACE DESIGN AND WORKFLOW

4.4.1 Authentication Interface

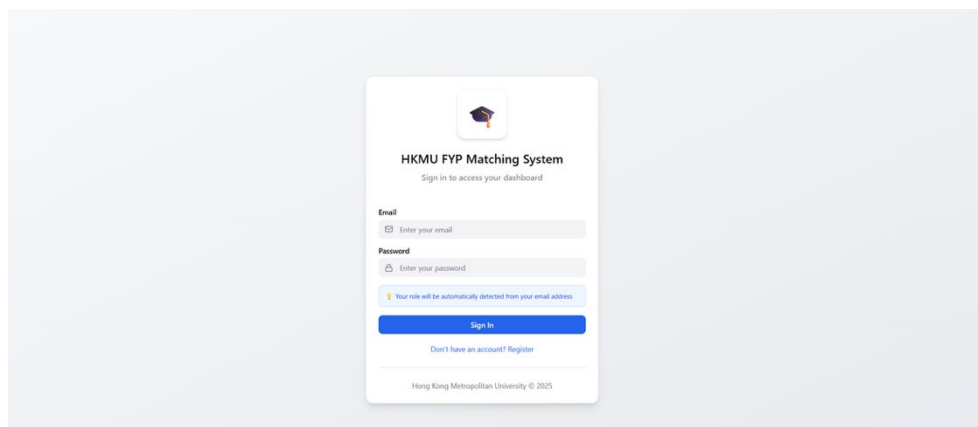


Figure 1 login page

In figure 1, the system has a single point of entry for students, teacher, and administrators. Upon logging in, you see fields where you can enter your email address and password. What I like about the system is its ability to identify your character based on your email address. After that, it takes you where you belong after logging in. There is also a link for signing up if you are not registered yet.

4.4.2 Student Portal Interface

The screenshot shows the 'Proposal Phase' of the HKMU FYP Matching System. At the top, there's a navigation bar with 'Proposal Phase', 'Matching Phase', 'Results', 'Popular Supervisors', and 'Categories'. Below this, a red banner indicates a deadline of 2025-03-20 23:59 to submit the project proposal. A 'Personal Information' dropdown menu is visible. The 'Student Profile' section displays the following details:

Field	Value
Name	John Doe
Student ID	12345678
Email	john.doe@hkmu.edu.hk
Major	Computer Science
GPA	3.75

Below the profile is the 'My Project Proposal' section, which includes a 'Project Proposal Form' with fields for 'Project Title', 'Description', and 'Required Skills'. A 'Submit Proposal' button is at the bottom.

Figure 2 student account, propose phase

Figure 2 contains an extensive proposal template wherein the student is required to fill out the title of their project, project description, and technology skills required for the development process. Information about the student profile such as their name, email, GPA, student ID number, and major are provided.

The screenshot shows the 'Matching Phase' of the HKMU FYP Matching System. At the top, there's a navigation bar with 'Proposal Phase', 'Matching Phase', 'Results', 'Popular Supervisors', and 'Categories'. Below this, a red banner indicates a deadline of 2025-04-15 23:59 to select project preferences. A 'Personal Information' dropdown menu is visible. The 'Project Popularity Stats' section displays a table of project options:

Popularity	Project Title	Supervisor	Selection Count	Action
🔥	AI Learning System	Prof. Li	23	<button>Add Preference</button>
⭐	Blockchain Application	Prof. Wang	15	<button>Add Preference</button>
📶	IoT Platform	Prof. Chen	12	<button>Add Preference</button>
○	Data Analysis	Prof. Huang	3	<button>Add Preference</button>
⭐	Mobile App Development	Prof. Li	18	<button>Add Preference</button>

Below the table is the 'My Preference Order (Drag to sort)' section, which currently shows 'No preferences added yet. Add projects from the table above.' and a 'Submit Preferences' button.

Figure 3 student account, matching phase

In figure 3, students are able to select projects from a live interface that illustrates how trendy each particular one is at that particular moment. Projects' names, supervisors, number of selected projects, and "Add Preference" button are included in the table. Moreover, students are allowed to rate projects and drag them into the preferred order, while the system informs about already selected projects.

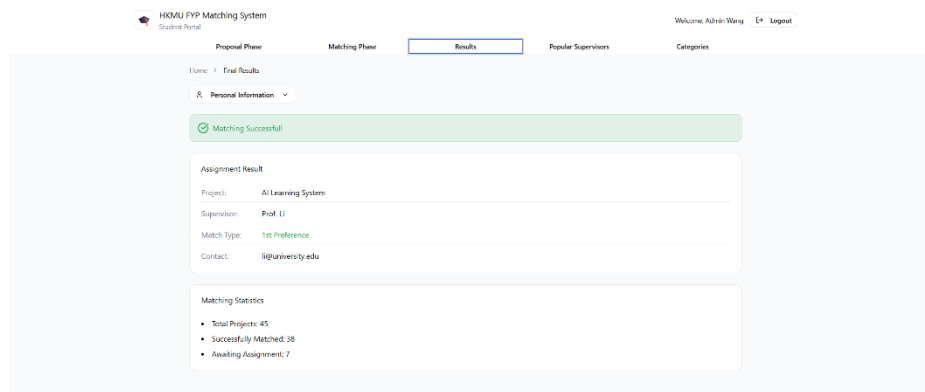


Figure 4 student account, result

After the matching process, it turns to result part just shown as figure 4. Students get assignment results that show their project, supervisor, match type ((like 1st Preference) and contact details. The system also shows matching stats (for example, 38 of 45 projects were a successful match) and links to helpful resources, such as popular supervisors and category stats.

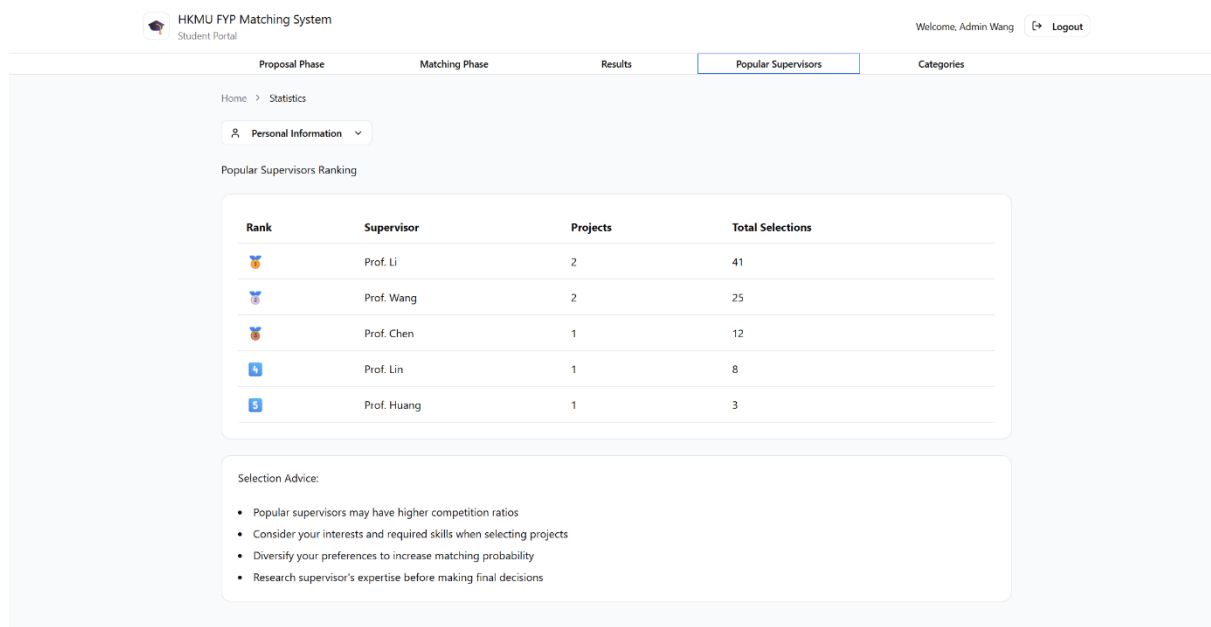


Figure 5 student account, popular supervisors

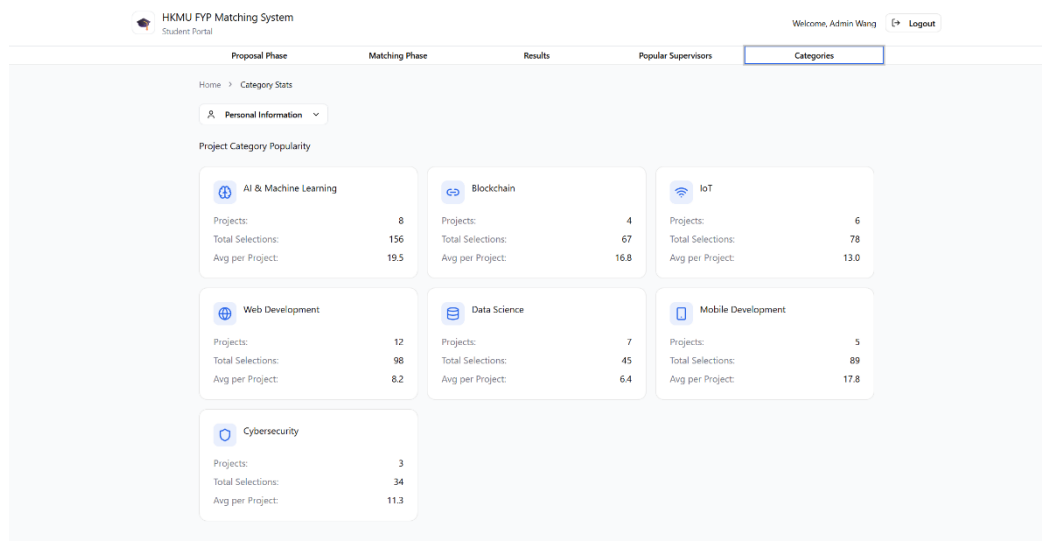


Figure 6 student account, categories

Figure 5 and 6 show how students can view supervisor rankings by project count and selections, plus category popularity stats. This information, along with selection tips, helps students pick projects that match their interests and have a decent chance of getting chosen.

4.4.3 APPLICATION MANAGEMENT INTERFACE

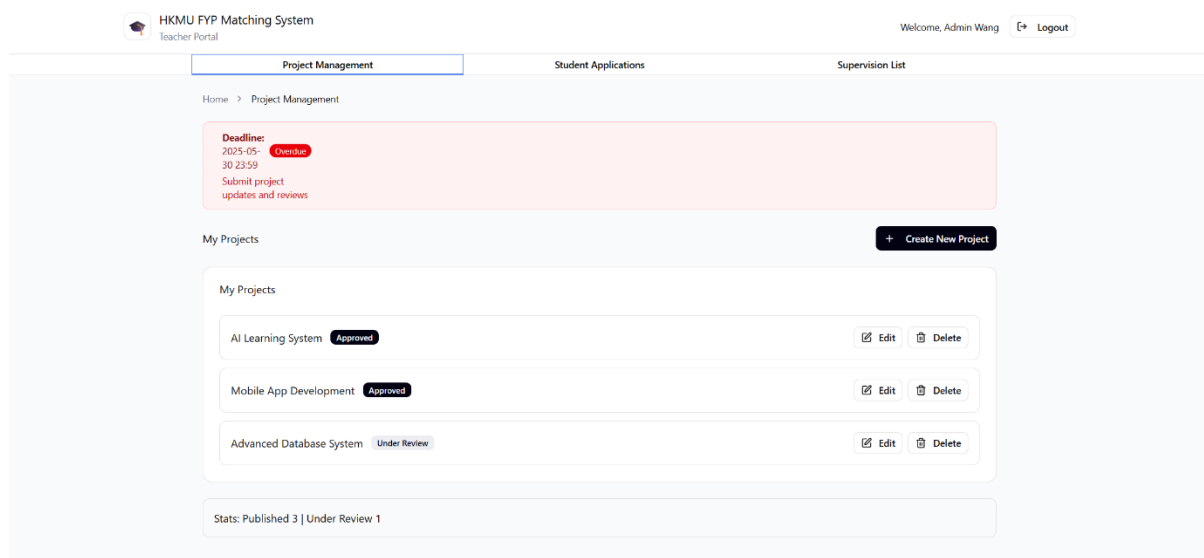


Figure 7 teacher account, project management

The supervisors have access to a dashboard where their respective projects and their corresponding lists of applicants are displayed as seen in figure 7 below. The supervisors

have the ability to view their applicants, including their GPAs and preferences, enabling them to make appropriate decisions.

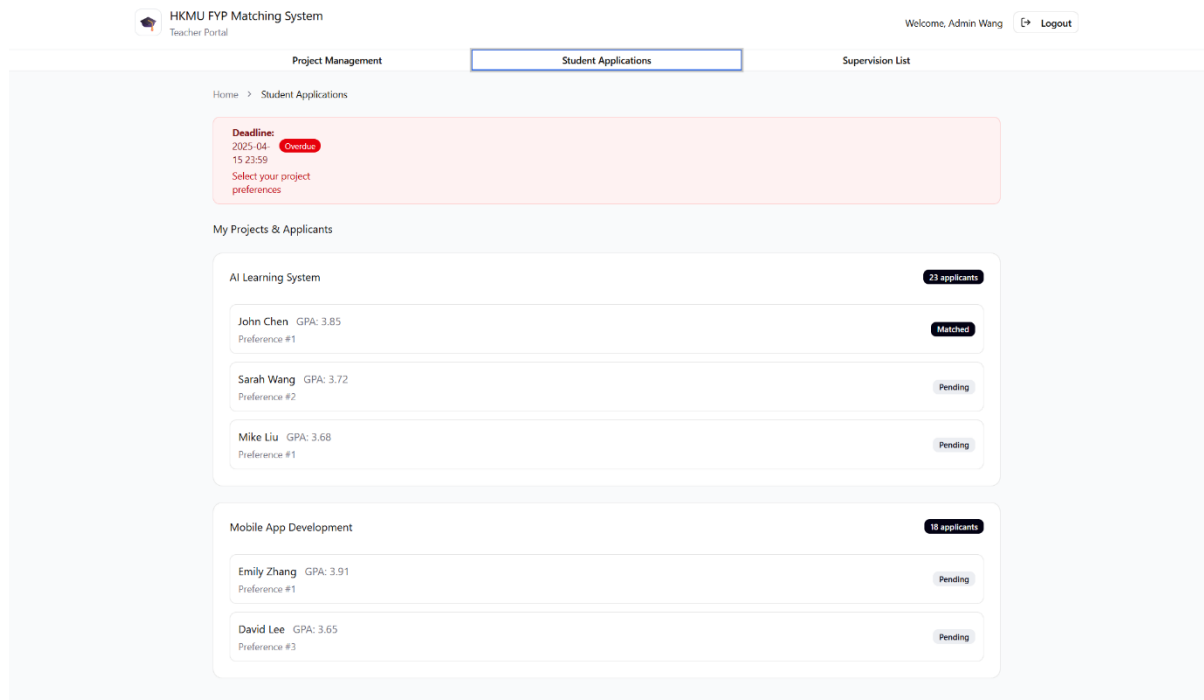


Figure 8 teacher account, student application

Instructors can view their assigned students along with their GPA, department, and contact details as seen in Figure 8. There are schedule management and meeting setup functions that assist in guiding projects throughout the year.

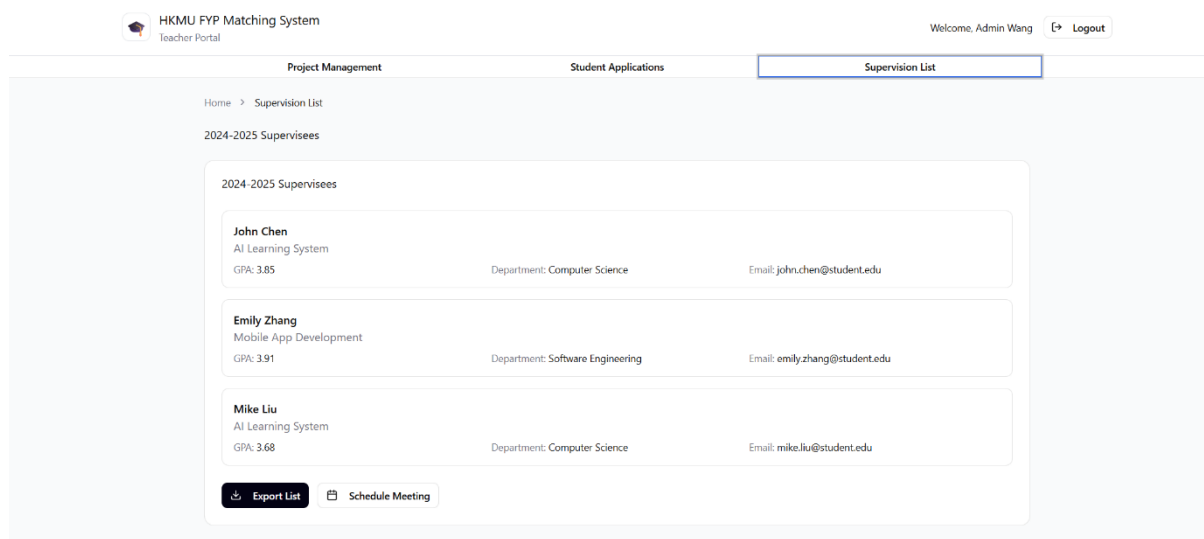


Figure 9 teacher account, supervision list

The Supervision List page represents the list of students assigned to each teacher for the particular academic year (e.g., 2024-2025), represented by figure 9. This is basically a simple

list that displays all the details of the assigned students including their name, the name of the project assigned to them, their GPA, department, and their email address. It is the ideal page to manage the supervision and includes features such as a Schedule Meeting feature.

4.4.4 Admin Portal Interfaces

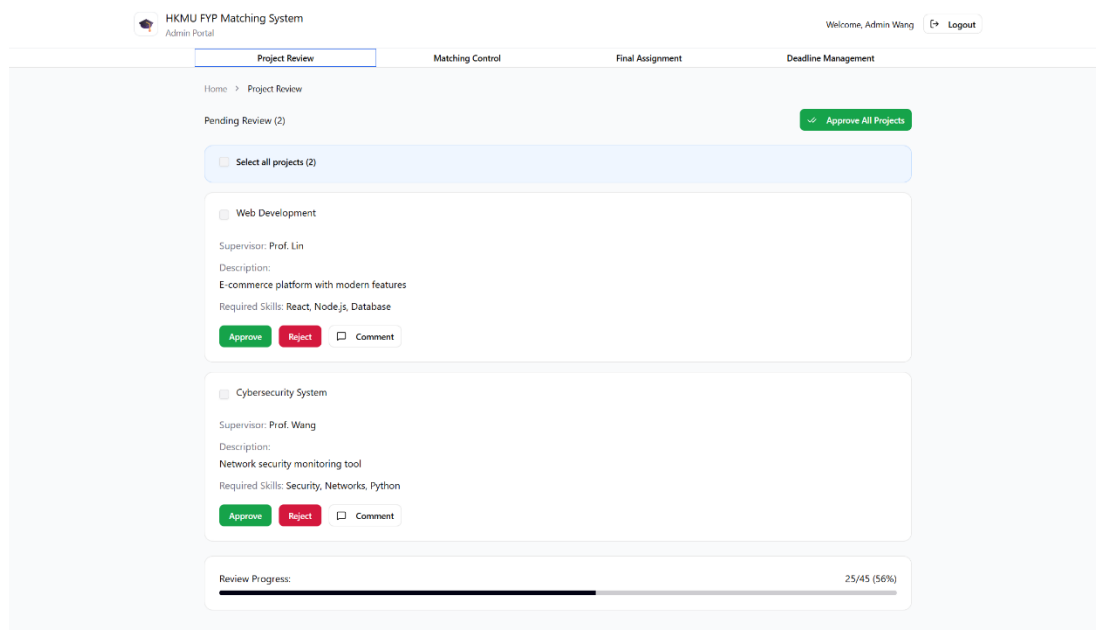


Figure 10 admin account, project review

The administrator can view the projects for which approval is required from the review dashboard as depicted in figure 10 below. The admin would have options such as viewing the details, necessary skills, approving/rejecting, or adding a comment for each project. There is also a status bar depicting the progress of the review process in terms of percentage.

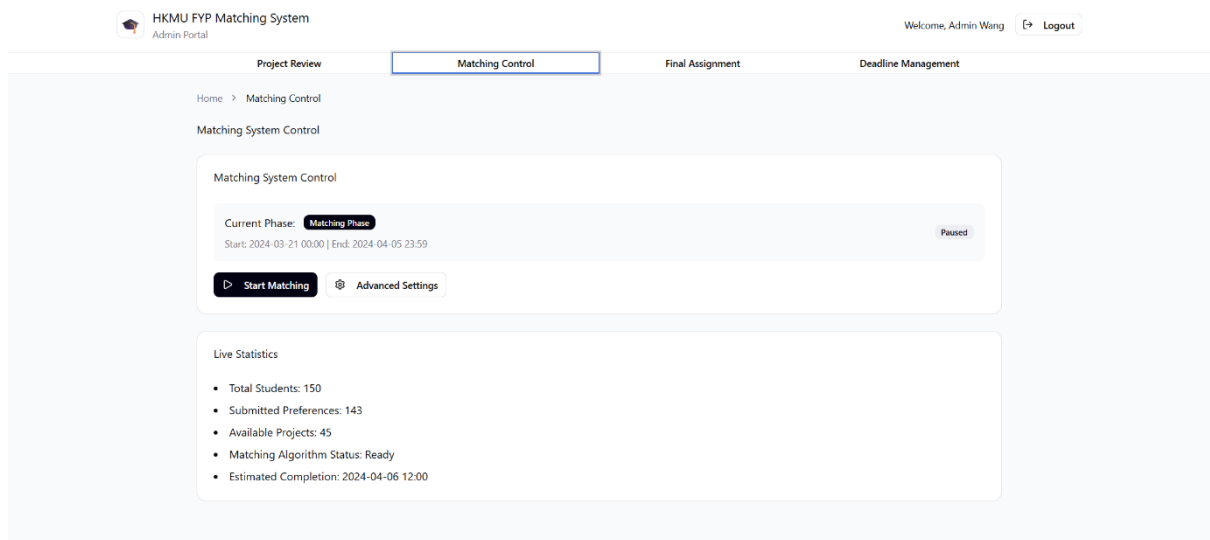


Figure 11 admin account, matching control

The admin dashboard of the matching system contains phase management using timestamping for beginning and ending phases, statistics such as number of participants (150 students), number of preference submission (143), and controls for algorithm execution, as illustrated in Figure 11 below.

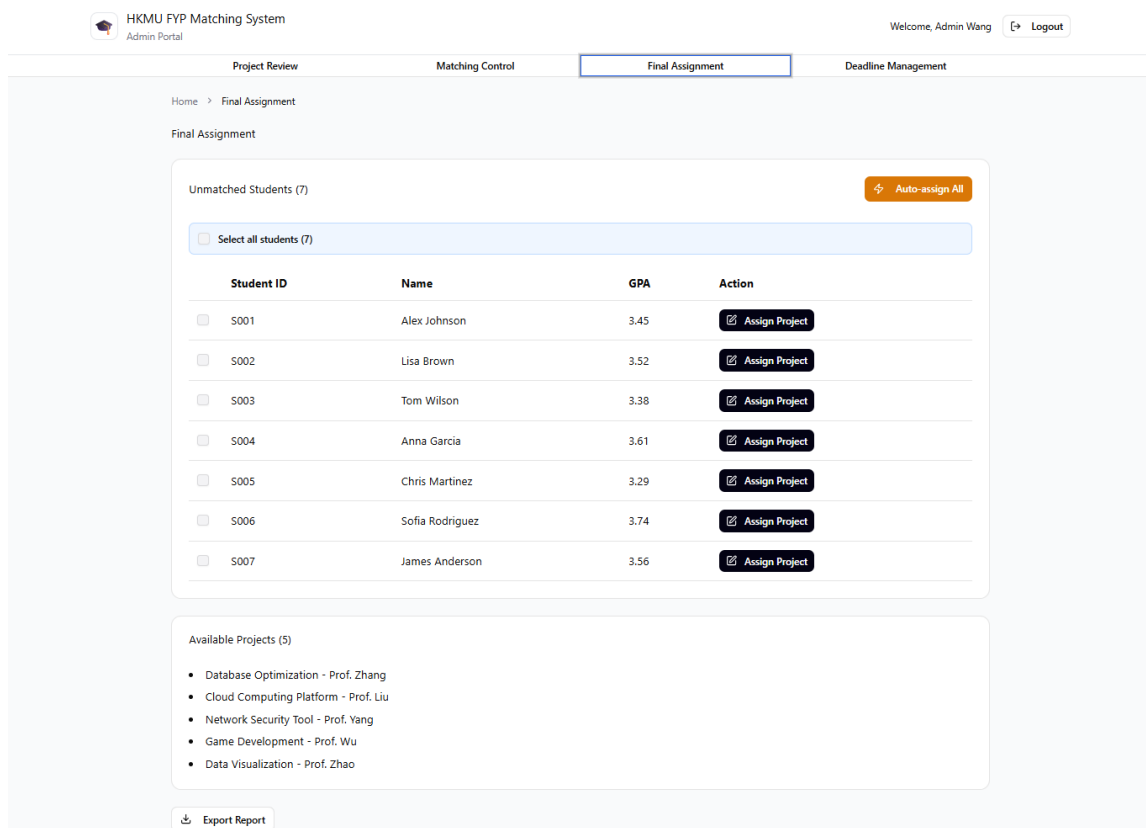


Figure 12 admin account, final assignment

However, where it does not automatically allocate students to projects, the process can be done manually, with just a simple click to match the remaining students to open projects. There is an indicator which gives a list of students who require projects allocation as illustrated in figure 12 below, including their GPA and open projects available.

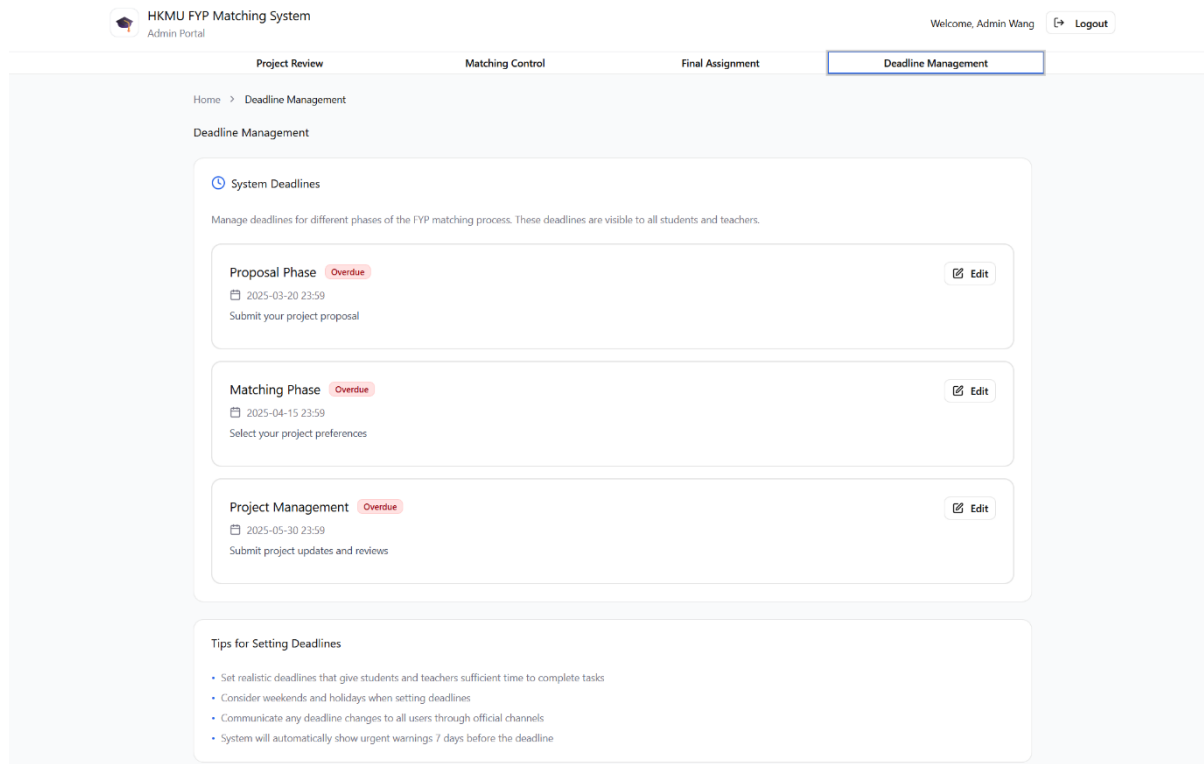


Figure 13 admin account, deadline management

Administrators are able to set deadlines for the entire system, modifying major phase schedules with automated reminders as indicated in figure 13 below. The administrator's interface manages all stages from the proposal process through matching to project management while considering academic calendars and communication policies.

4.5 SYSTEM WORKFLOW SUMMARY

Stage 1: Proposal and Review

Students propose ideas for projects, and then administrators confirm and approve project lists. With this, the application system is responsible for ensuring that the process of reviewing is orderly and all project requirements, such as tracking and maintaining high standards, are being met.

Stage 2: Preference-based Matching

Students browse for projects of interest, see which projects are in demand, and then prioritize those projects. As a result, an advanced algorithm matches each student to a project, taking into account preferences, academic performance, and number of slots available in each project.

Stage 3: Reporting and Project Management

Students receive an approved list of project matching together with the information about their supervisors' contacts. Teachers, in turn, receive their students' lists. Administrators are responsible for solving individual problems and manual matching. They also manage projects by applying deadlines.

Such interface is appropriate for all parties concerned, since it gives them necessary instruments depending on the task and is user friendly [9]. This is achieved due to the combination of automation and administrator control, which means that workload distribution is reasonable.

4.6 DESIGN AND IMPLEMENTATION UPDATES

The following section highlights the key changes and enhancements made to the design since the initial proposal. This section is concerned with the differences and additions that have been made to the proposed design.

4.6.1 Technology Stack Enhancements

As described in Section 2, the reason for using the MERN technology stack was given. In the meantime, certain changes have been made to make the process simpler, more structured, and more efficient.

Table X provides a comparison between the proposed high-level architecture and the implemented system:

COMPONENT	ORIGINAL DESIGN	IMPLEMENTED SOLUTION
Frontend framework	React (generic)	React 18 with Vite 5, providing a faster development server,

		optimized builds, and a modern tooling ecosystem.
Styling approach	Inline styles / global CSS	Modular CSS with per-component stylesheets, improving maintainability and reducing style conflicts across pages.
State management	Not explicitly specified	React Hooks (including useState, useEffect, and useMemo) for reactive state management and performance optimisation.
Routing	Not explicitly specified	React Router DOM v6 with role-based protected routes for Students, Supervisors, and Administrators.

Table 3

These modifications maintain the existing three-layer framework while simplifying code modification and extension. Vite provides instant server start and lightning-fast Hot Module Replacement (HMR) that improves the development experience [18].

4.6.2 Modifications to User Interfaces

User interfaces remain similar to their original versions depicted in Section 3.1; however, several usability enhancements have been implemented to ensure the smooth operation of the system.

4.6.2.1 Drag-and-Drop Functionality for Ordering Options

While the initial design featured ranking of the project options in a straightforward manner, students now can rank their preferences via drag and drop. This approach ensures the convenience of ranking while avoiding errors in reordering via button presses.

4.6.2.2 PROJECT FILTERING

The functionality of the project search tool has been enhanced to aid students' choice of projects. Improvements include the following: searching for projects based on keywords from titles, descriptions, and supervisors; filtering projects by required skills through tags extracted from the list of required skills of every project; selecting projects by their supervisors through a drop-down menu; and sorting projects according to their popularity, title, and supervisor's name.

4.6.3 Matching Algorithm Enhancements

The initial approach was to adopt the most basic, easy-to-explain matching approach which would be based mostly on preferences of the students and their grades. At this point, however, this approach will look like Gale-Shapley matching but with additional restrictions and tie-breaking rules applied.

What the improved algorithm includes is as follows:

- GPA Tie Breaker - should there be more students who prefer a certain project, select among them the one with higher GPA.
- Limits on the number of people working on each project.
- Limits on the number of projects chosen by each student - no more than 10 projects.
- Limitation on number of choices per student - ordered list of at most 10 projects.
- Tracking popularity of each project based on number of students interested.

All the above-mentioned data will be available in the Students Portal.

This information shows up in the Student portal. These updates make the whole process fairer and easier to understand, while still sticking to the original plan [15].

4.6.4 Matching Algorithm Implementation Details

The matching algorithm forms the backbone of the FYP Matching System, and the objective of this algorithm is to match students fairly to their projects depending on their preferences and performance. The following part explains how the matching algorithm works.

Algorithm Overview

This algorithm uses multiple steps in assigning projects to students while taking into account their preferences but using the students' GPAs as a tie-breaker mechanism in case projects get over-subscribed. The algorithm assigns projects to students starting from those with the highest GPAs.

Algorithm Process

The initial stage starts off with the ordering of all the students who have made submissions of their preference list. These students are arranged in the decreasing order of their GPA, while the students who have not submitted any preference are excluded from the matching stage and are to be considered for the clearance stage.

The second step involves the iteration over each individual student. For each student, the ranking of his/her preferences are analyzed in the order of priority. The number of projects for which the student may rank is ten. In analyzing any particular preference of a student, the first step is to verify if there are any available slots in that project.

In case where the project capacity is full, the algorithm checks whether the GPA of the student under consideration is greater than the GPA of the current student assigned to the project. If yes, then the current student is replaced by the student under consideration, and he is now unmatched.

Thirdly, the algorithm looks out for all the unmatched students.

Tiebreaker Mechanism

In cases where multiple students compete for the same project and have identical GPAs, a tie-breaking condition based on an additional parameter is implemented to achieve determinism in the results. The system considers the timestamps at which students submitted their preferences with respect to the projects and those that submitted first get prioritized. This guarantees consistency and determinism of the results.

Popularity Tracking

To help students rank their preferences by taking into account the level of popularity of different projects among other factors, a dynamic counter of popularity is maintained by the system. Each time a student adds a project to his/her list, the counter gets incremented, and when the project is deleted, the counter is decremented.

Time Complexity

The overall complexity of the matching algorithm is $O(n \log n + nk)$. Here n represents the number of students, and k represents the average number of preferences for one student. It consists of two stages, namely sorting with $O(n \log n)$ and matching with $O(nk)$ time complexity.

4.7 DATA EXPORT FUNCTIONALITY

Some CSV export options are being added on the server. This makes it easier to keep records, create reports, and do other analysis. The data can export out of the system.

4.8 AUTHENTICATION AND ACCOUNT MANAGEMENT

This system makes use of an authentication method that is secure and incorporates mandatory password management starting from the time of account creation.

During account creation via batch creation by the admin, this system assigns temporary passwords according to institution naming conventions. Passwords for students begin with the name "Changeme123!," and for teachers, the first eight passwords will be from 00000001 to 00000008. All newly created accounts have the `mustChangePassword` attribute automatically set to true.

Once the account owner logs into the portal for the first time, he or she is immediately transferred to the password change page when the `mustChangePassword` attribute is true. There is also a mandatory password length of eight characters and no use of `initialPassword` as the password, ensuring that no one will merely confirm his or her default password.

Authentication also facilitates the changing of normal passwords at any time for users on both portals.

4.8.1 Batch Account Creation via CSV Import

As a way of simplifying the process of creation for large numbers of users, batch creation of user accounts is facilitated. The batch creation may be carried out using either individual form completion or through the upload of a CSV file.

The account creation page for admins has two modes of operation. The first mode is the single-entry mode, where the admin fills out the form with the information about students or teachers. The second mode of operation is the batch creation mode, where an administrator can download a CSV file, fill out it with data and then upload the data back to the application.

The CSV file that needs to be filled should have four columns which are: Student ID, Name, Major, and Notes for students and Teacher ID, Name, Department, and Major for teachers. The application checks whether the required columns exist in the CSV file and the values in the required columns are valid. It ensures that only ECE, CCS, and ECE +CCS majors are allowed. Once it finds valid data, it creates accounts in batches while setting the required default credentials for accounts.

4.8.2 Student Self-Proposal Feature

This system offers students the opportunity to propose topics on their own as part of the FYP topic proposal process. This helps students with particular research areas to get supervisors' approval for developing unique project proposals.

When students propose topics independently, they use the portal to submit the proposal using the proposal submission form. The form requires students to input the title of the project, a detailed description of the topic that should not be less than fifty characters, as well as the skills required for the project from the pre-selected list, and additional skills.

Upon submission, the proposal moves into the review phase. Matching major specialists among teachers can access the proposal on their portal and approve or reject it. In case a teacher

decides to approve a proposal, the system automatically allocates that teacher as the project supervisor and marks the project proposal as approved. If the proposal is rejected by the teacher, the system automatically marks it as rejected, notifying the student about it.

In case a student fails to get the proposal approved within the self-proposal period, the system automatically rejects all such pending proposals via a scheduled task. The status of a proposal can be viewed by the student from the portal interface.

4.8.3 Drag-and-Drop Preference Ranking

The My Preferences module features an easy-to-use drag-and-drop design that helps students rank the projects according to their preferences.

Students may select projects and add them to their preference list using the available projects and clicking on an Add to Preferences button. Projects added to the list will then be displayed according to their rankings, and users can rearrange them by moving the items up or down using their mouse or clicking on Up/Down buttons to move them one position at a time.

In terms of functionality, the drag-and-drop feature leverages the HTML5 drag-and-drop capabilities. During the reordering process, visual cues are presented to identify the draggable item and its destination. As changes are made, they are immediately saved and synchronized with the backend server, so that the preference ranking is retained even after refreshing the web page.

To avoid analysis paralysis among users, there is a limit of ten preferences per student. After adding the preferred projects to the list, students can submit their selections. After submission, the preference list will be locked for editing, and no further modifications will be allowed.

4.8.4 Real-time Assignment Status Polling

Once the matching process is completed, the students should be made aware of the time they can know the result of their allocation. In this system, the use of real-time polling is done for checking updates in the matching process.

On the Results screen, the student's browser makes polling calls to the server after each five-second interval to see if any assignment has been made by the matching process. If an assignment is made, the polling process is stopped and the system displays the results of the matching process, including the title of the project, supervisor's name, and the code of the project. The system also uses the banner feature while the matching process is going on or there are no results yet.

4.8.5 Major-Based Project Filtering

Students may apply filters based on their academic major and view projects based on this criterion. It should be noted that there are three major groups that are recognized by the system, namely, Electronics & Computer Engineering, Computer & Cyber Security, and the combination that allows projects to take in students from both majors.

While viewing projects, students apply filters according to their major, and those projects that match the specified major will appear in the list. Teachers, while developing projects, indicate the majors that the project is open for. Proposals submitted to teachers are seen only from the perspective of students that study in the major that is similar to the one indicated by the teacher.

5. EXPERIMENT

5.1 SYSTEM IMPLEMENTATION OVERVIEW

The current prototype has been implemented using React 18 with Vite on the frontend and Node.js/Express with MongoDB on the backend. The main setup, login system, and Student portal have been completed according to the original plan. Role-based portals for students, supervisors, and administrators have been set up and are running in a local development environment. The Teacher and Admin portals currently provide their main pages and layouts, with further backend integration planned.

5.2 MATCHING ALGORITHM IMPLEMENTATION

In the current implementation, the matching algorithm is based on a Gale–Shapley-style design with additional constraints.

5.3 SCREENSHOTS

5.3.1 Login Portal

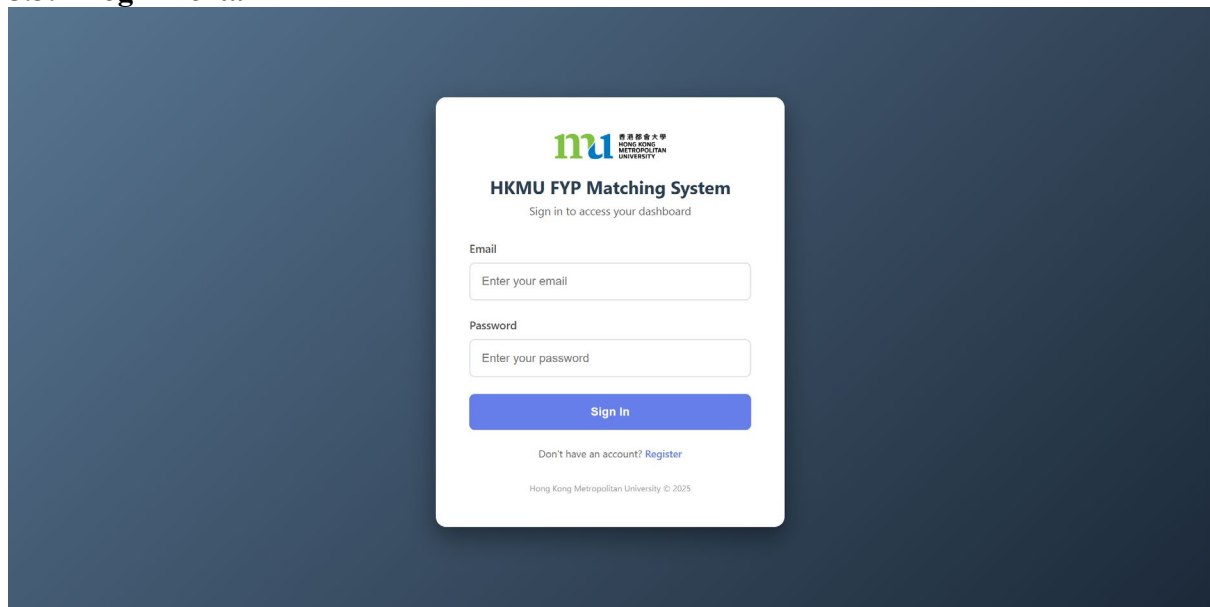


Figure 14 login interface

It shows the implemented login interface, which provides a single entry point for all user roles. After entering their email and password, the system identifies the user role and redirects them to the appropriate portal.

5.3.2 Student Portal

The screenshot displays the 'Submit Proposal' form within the HKMU FYP Matching System Student Portal. The left sidebar contains navigation links: Proposal, Browse Projects, My Preferences, Results, and Profile. The user's name, Chan Tai Man, and GPA, 3.45, are shown at the bottom of the sidebar. The main form area is titled 'Submit Proposal' and includes a sub-header 'Fill in the details below to submit your proposal'. The form fields are: 'Proposal Title' (text input), 'Description' (text area with a minimum 50 characters requirement), and 'Required Skills' (checkboxes for Python, JavaScript, Java, C/C++, Machine Learning, Deep Learning, Web Development, Mobile App, Data Science, Cloud Computing, IoT, Cybersecurity, Blockchain, and Other). At the bottom of the form are 'Submit Proposal' and 'Clear Form' buttons.

Figure 15 proposal form

Here, students can propose their own FYP topics by entering the project title, description and required skills.

The screenshot displays the 'Browse Projects' page within the HKMU FYP Matching System Student Portal. The left sidebar contains navigation links: Proposal, Browse Projects, My Preferences, Results, and Profile. The user's name, Chan Tai Man, and GPA, 3.45, are shown at the bottom of the sidebar. The main content area is titled 'Browse Projects' and includes a search bar, a 'Sort by Popularity' dropdown, and a 'Filter by Skills' dropdown. Below the filters, there are four project cards, each with a title, supervisor name, description, required skills, and 'Add to Preferences' and 'View Details' buttons. The projects are: 'Cybersecurity Dashboard for Small Businesses' (Supervisor: Alex), 'Social Engineering Awareness Tool' (Supervisor: Alex), 'Blockchain-Based Voting System' (Supervisor: Alex), and 'Mobile Application Security Testing Tool' (Supervisor: Alex). Each card also shows '0 selections' in a red badge.

Figure 16 browse projects page

Here, students can look through all the available projects by searching keywords, filtering by skills, and filtering by supervisor. You can see the project title, supervisor name, a brief description, needed skills, and how many students have selected the project on each project card. From here, students have the option to add projects to their preference list.

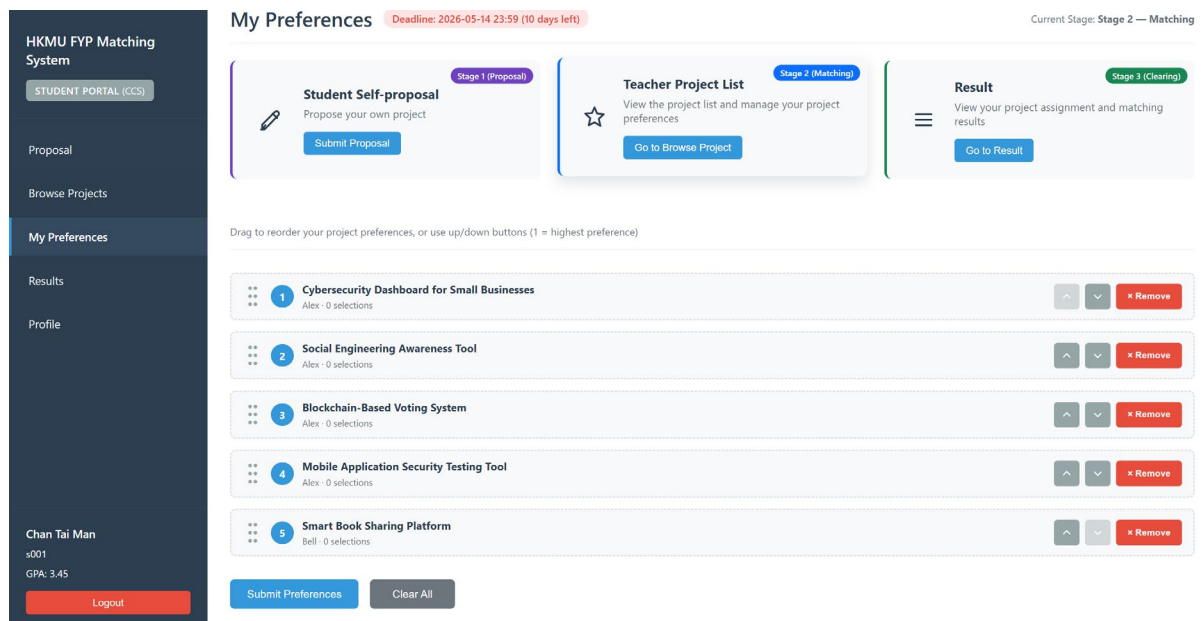


Figure 17 my Preferences page

Students can manage their project preferences. When a project is added from the browsing page, it will appear here. Students can change the order of their choices by dragging and dropping, use the up and down buttons to fine-tune the order, remove projects from the list, clear all preferences, and submit their final ranked preferences when they have finished.

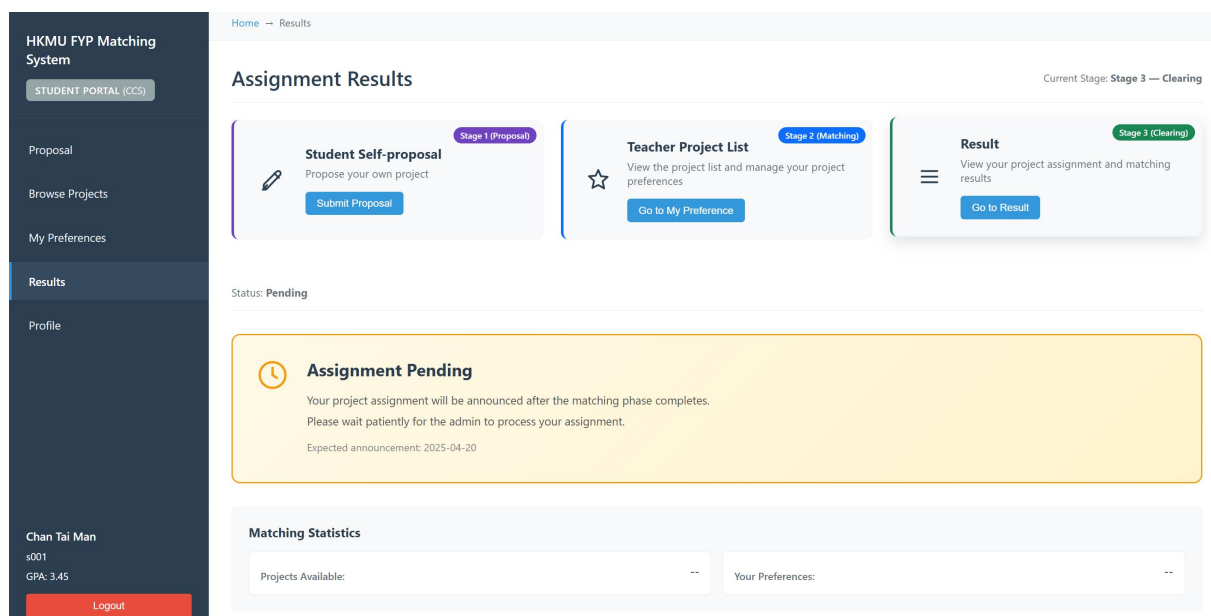


Figure 18 student's matching results page

It shows the final project and supervisor assigned to a student after matching. Information such as the student's preference and supervisor contact details are listed. This page will also show statistics for context.

Teachers can check out FYP proposal from students, as well as the title, what it is about, the skills students need, and the student's details. This lets teachers choose which student projects they want to guide.

[illegible]

Page 43

Here, supervisors can handle suggested topics from themselves and view approved/rejected students proposal.

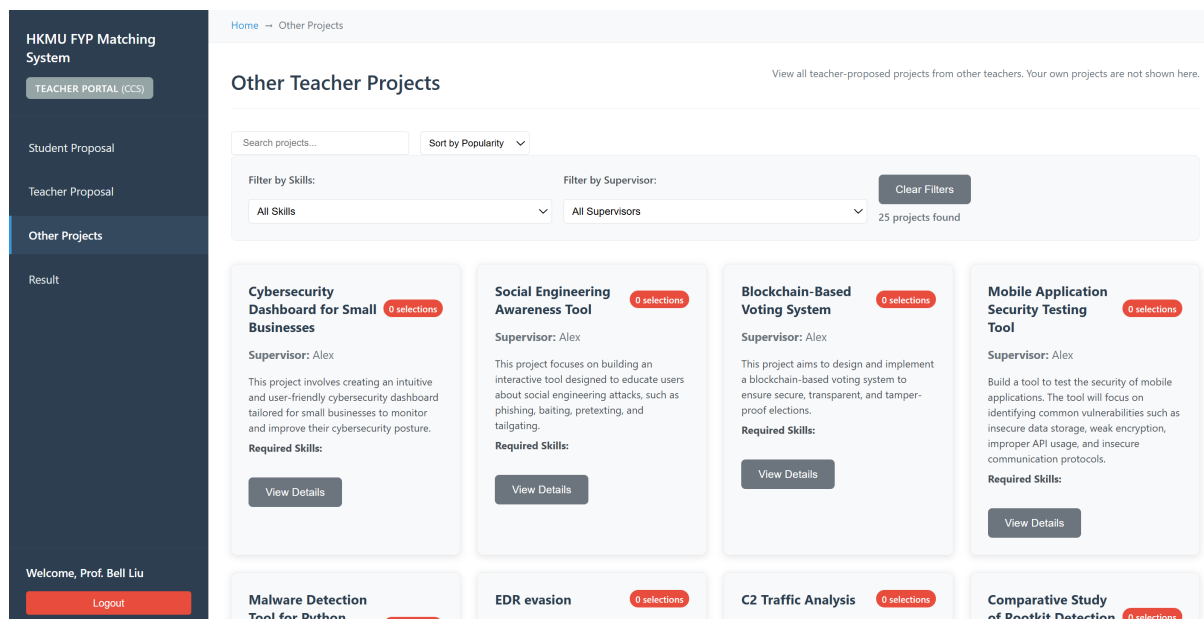


Figure 23 other project page.

Teacher can browse other teacher's proposal over this page, filter function can also be used. For teacher's reference when they are creating projects.

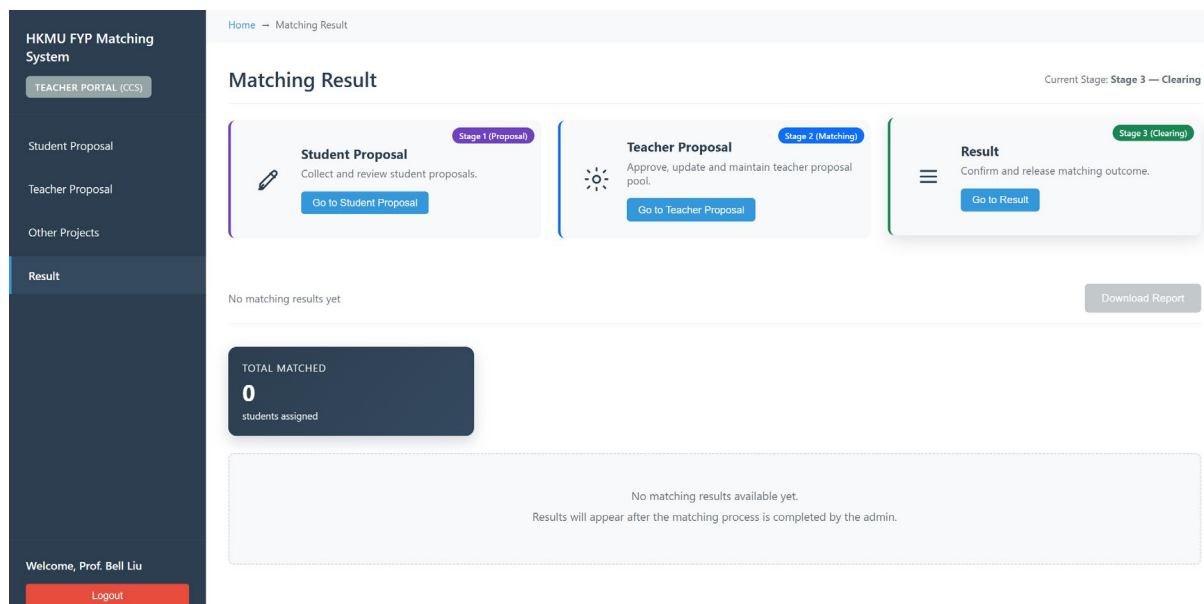


Figure 24 supervisor's matching results page.

In this section, instructors will find out about the final outcome of the matching for their projects. They will know who is included in their teams and will be provided with information

about the matching procedure, along with the ability to show or hide the matchings. Instructors will also be able to export reports on the matchings.

These pages showcase how the Teacher portal gives instructors tools to organize projects, evaluate student proposals, assess matching results, and keep track of supervisees.

5.3.4 Admin Portal

The screenshot displays the 'Matching Control' interface of the HKMU FYP Matching System. On the left is a dark sidebar with navigation links: 'ADMIN PORTAL', 'Matching Control' (active), 'Create Account', 'Deadline Management', 'Final Assignment', and 'Data Export'. The main content area has a breadcrumb 'Home → Matching Control' and a title 'Matching Control'. Below this is a 'Matching System Control' section with a 'Current Stage' indicator showing a timeline from 2024-03-21 to 2024-04-05, and three buttons: 'Start Matching' (blue), 'Reset Server' (grey), and 'Refresh Stats' (grey). The 'Live Statistics' section shows 'Matched Projects: 0', 'Available Projects: 30', and 'Matching Algorithm Status: Ready' (green). A 'Statistics by Major' table follows, showing data for ECE and CCS majors. At the bottom, a summary states 'Total: 50 students, 30 projects, 0 matched'. The sidebar footer includes a welcome message for 'Admin Wang' and a 'Logout' button.

Major	Students	Projects	Matched	Remaining
ECE	25	30	0	25
CCS	25	0	0	25

Figure 25 control page

From here, administrators can set up and run the matching process. The page shows key numbers such as the number of students, submitted preference lists, and some information about the current stage. After starting the matching process, the status and key stats are shown so administrators can follow its progress.

Figure 26 create account page

Account for student and teacher can be created manually on 'create account' page. Manual creation involves entering data manually or by importing data from Excel files. In case of new account creation, a default password will be set (Changeme123!).

Figure 27 deadline management page

Admins can set and manage dates for each part of the Final Year Project, like when to turn in project ideas, pick choices, and pair up. These dates show up on the home pages of both student and teacher sites, so everyone knows where things stand and when things are due.

HKMU FYP Matching System

ADMIN PORTAL

Matching Control

Create Account

Deadline Management

Final Assignment

Data Export

Welcome, Admin Wang

Logout

Unmatched Students (50)

☐ Select all (50)

Student ID	Name	GPA	Action
☐ s001	Chan Tai Man	3.45	Assign Project
☐ s002	James Ng	2.7	Assign Project
☐ s003	Emily Smith	2.68	Assign Project
☐ s004	Daniel Robinson	3.54	Assign Project
☐ s005	Sophie Cheung	2.92	Assign Project
☐ s006	Ryan Brown	3.5	Assign Project
☐ s007	Chloe Clark	2.65	Assign Project
☐ s008	Kevin Lam	3.4	Assign Project

Prev 1 / 5 Next

Selected: 0 / 50

Matched Students (0)

Available Projects (30)

Auto-assign (0 selected)

Cybersecurity Dashboard for Small Businesses (D11)
- Alex Teacher Proposed 0/1 assigned

Social Engineering Awareness Tool (D12)
- Alex Teacher Proposed 0/1 assigned

Blockchain-Based Voting System (D13)
- Alex Teacher Proposed 0/1 assigned

Mobile Application Security Testing Tool (D14)
- Alex Teacher Proposed 0/1 assigned

Smart Book Sharing Platform (L11)

Showing 5 of 30

Export Report

Figure 28 final work assignment page.

It shows the final matching results and lists students who were not matched by the system. Staff can then assign these students to the right projects by hand, or they can use the auto-assign choice to fill the open spots. The page also lets you save the final report as a CSV file.

HKMU FYP Matching System

ADMIN PORTAL

Matching Control

Create Account

Deadline Management

Final Assignment

Data Export

Welcome, Admin Wang

Logout

Home → Data Export

Data Export

Export system data in CSV format for analysis and reporting

Project List
Export all available projects with details and requirements
Download Project List

Student List
Export complete student information including preferences and assignments
Download Student List

Teacher List
Export all teacher information including contact details and majors
Download Teacher List

Matching Results
Export current matching results showing student-project assignments
Download Matching Results

Export All Data
Download all four reports at once for comprehensive data analysis
Download All Reports

Export Information

- All files are exported in CSV format for easy import into Excel or other spreadsheet applications
- Matching results show current student-project assignments based on preferences
- Student list includes GPA, major, and submission status
- Project list contains capacity, popularity, and requirement details

Figure 29 data export page

This is where administrators can export different types of data. This includes matching results, student lists, teacher list, project lists, or everything combined into one export all dataset. These CSV files can be used for later analysis, auditing, and backups.

The Admin portal screenshots show how administrators can manage the FYP allocation. They can control project approval, run the matching process, adjust final assignments, handle deadlines, and export important data, giving them complete control.

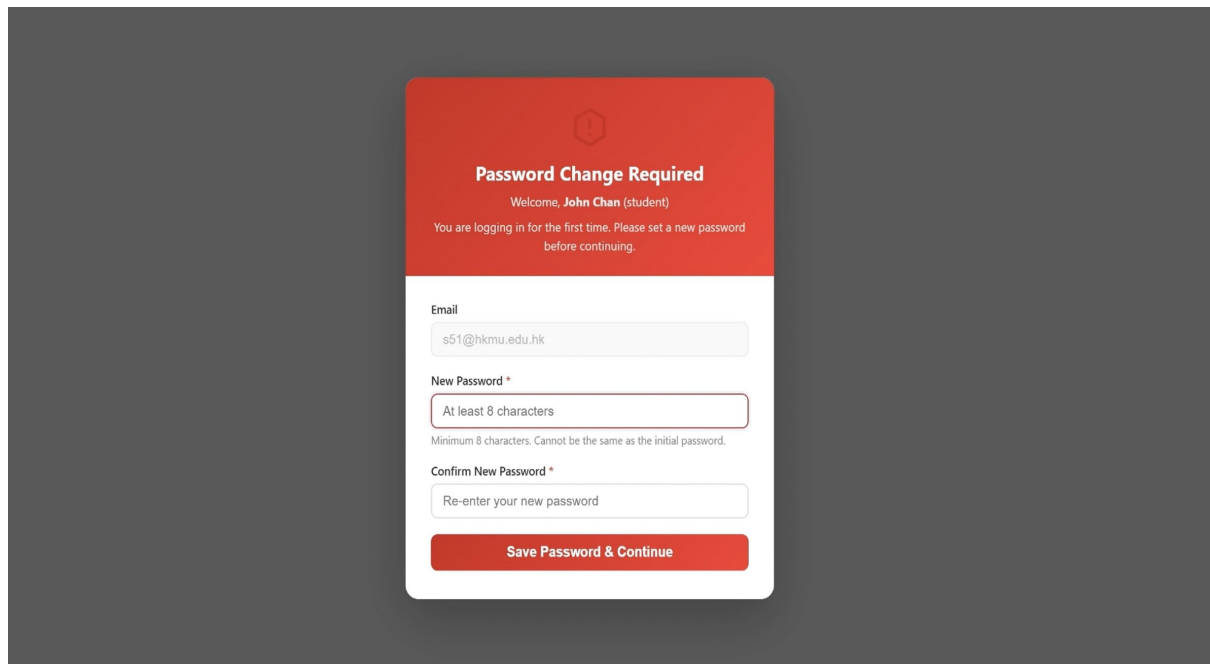
The image shows a login interface for a new account. At the top, a red banner contains a shield icon with an exclamation mark, the title "Password Change Required", and a welcome message: "Welcome, John Chan (student)". Below this, a message states: "You are logging in for the first time. Please set a new password before continuing." The form below has three input fields: "Email" with the value "s51@hkmu.edu.hk", "New Password *" with a placeholder "At least 8 characters" and a note "Minimum 8 characters. Cannot be the same as the initial password.", and "Confirm New Password *" with a placeholder "Re-enter your new password". A red button at the bottom is labeled "Save Password & Continue".

Figure 30 new account login page

When the first time user login to the new created account (student/teacher), system require user to change the default password to their own password. Make sure it is secured for all students and teachers.

6. TESTING METHODOLOGY

This chapter details the test plan that will be used to make sure that the system is reliable, stable, and secure.

6.1 TESTING STRATEGY OVERVIEW

The testing process is hierarchical. Input validation and error handling filter out any erroneous data that enters the system at the API level. Functional testing ensures that all functionalities meet the users' needs. Integration testing checks if each part works well with other parts. System testing checks if the whole system works well as a unit [17].

6.2 FUNCTIONAL TEST SCENARIOS

6.2.1 Authentication Tests

Testing has to be conducted for several test cases that will involve successful logging in, failed login attempts, first-time login and password change, password change for users who need to change their password, and accepting valid passwords. Login test cases include logging in using valid credentials; the login attempt should be successful, and the portal that appears is determined by the user type. Logging in with invalid credentials should return an error message on the same login page. When logging in for the first time, if `mustChangePassword` returns true, then the next page should be password change. The password change tests include checking for minimum password lengths, not allowing use of the same password for the old password, accepting valid new passwords, and updating user details.

6.2.2 Student Portal Tests

Project browsing includes listing all projects based on the student's specialization, searching for projects using keywords in title, descriptions, and supervisor names, listing only projects that match skills chosen, adding projects to favorites, removing projects from favorites, rearranging project preferences using drag and drop, locking preferences after submitting them,

and validating descriptions according to the minimum length requirement. Submission of proposals should also be tested to check if teachers review proposals.

6.2.3 Teacher Portal Tests

Tests are required on the project management and proposal review in the teacher portal. The creation of a project needs to assign the proper type and supervisor. For reviewing the proposal, it is needed to show only those proposals whose major is same as that of the teacher. Approval of proposal should update the project status and assign the teacher as its supervisor. Rejection of proposal needs to update the proposal status. Supervision list should properly show the assigned students.

6.2.4 Admin Portal Tests

It is required to test the following functions of the admin portal. Updates in deadline should update the timestamp. Starting the matching process needs to start execution of the algorithm and create the record of assignments made. Resetting of preferences will reset all preferences and assignments made. Batch creation of account will process the csv file and create user accounts. Manual assignment will create assignment records. Exporting CSV will generate the proper csv files.

6.2.5 Integration Test Scenarios

Integration testing ensures that end-to-end flow occurs throughout multiple modules.

The correct sequence for matching flow should be the following. Teachers generate projects in the proposal period. Students make proposals according to preference before the deadline expires. The match engine starts operation from the administrator side. Students see what projects have been assigned to them. If there are any unmatched students, the administration matches them manually.

The student proposal flow should be the following. The student proposes a project. The proposal will appear in teacher review queues of the corresponding major. The teacher accepts

the proposal. The project will become available to all students. The student proposing the project sees it accepted.

6.2.6 Edge Case Testing

Edge cases need to be accounted for in the design of the system. When no preferences are offered by the student, the matching engine successfully flags the student as being unmatched. In a situation where there is no place left to allocate other projects, the system continues evaluating other proposals. When a teacher deletes an assigned project, these preferences are not taken into account by the matching engine.

7. RESULTS AND DISCUSSION

7.1 PROJECT STATUS AND ACHIEVED MILESTONES

The project began with phases on setting up the system, designing the portal, developing the algorithm, and testing. By the time of writing this report, the setup process, login system, and basic Student portal have almost been completed as per the plan. The algorithm has also been developed ahead of the scheduled timeline. On the other hand, the development process of Teacher and Admin portals is ongoing, which needs some work on the back end. Visualization of data, testing systematically, and preparations for deployment have not yet begun and will be done in the coming phase.

8. CONCLUSION AND FURTHER WORK

8.1 CONCLUSION

The motivation, design, and status of development of the FYP matching system have all been presented above. The purpose of this system - to provide an easily-accessible and centralized online system that matches students with their projects - remains as before. A web application that is designed using role based access control with sections for the students, supervisors, and administrators has been developed. This has enabled the improvement in the matching algorithm such that students can be assigned projects according to their preferences, GPA, and capacity limits of each project. Though the system cannot be deployed as yet, the progress achieved thus far proves that the proposed system is feasible and achieves its objectives.

The implementation of the FYP Matching System is one instance where the use of the web development technology comes handy in solving administrative challenges when allocating students projects. It is important that the transparency factor is incorporated in the design of this project because this will help solve the lack of transparency challenge that was identified by the students.

The need for a mandatory password change by users after the initial login has been recommended as the best practice since it will help replace the default login credentials with the actual credentials before carrying out any other activity. It will save time by generating a number of accounts for all the users simultaneously.

The drag and drop feature in allocating preference will be useful in preventing mistakes that were experienced by the students when manually allocating their preferences. Together with real-time popularity tracking, this feature will enable the students to make the right decisions as they select their projects.

When it comes to allocating students projects based on their GPAs, the algorithm considers the students and matches them according to that order. This process of tying breaking that takes place in this system is transparent.

Alternative ideas in regard to future developments would be a notification system, which will alert the user about important milestones like approaching deadlines, approved proposals, and

successful match-making. Further additions might include a way to visualize other data, for example the most popular project and number of successful matches created. Lastly, an alternative might be using the single sign-on option offered by the platform for various institutions.

8.2 DATA DISPLAY AND BASIC ANALYSIS

At present, basic data such as popularity scores are provided to assist students with decision-making. The next step would involve a simple graphic interface to provide users with information regarding the options of the project selection process. Possible future developments include the following:

1. The addition of a small statistics section displaying total student count, project count, and total number of student-project matches.
2. Basic graphs representing the popularity of projects (e.g., bar graph illustrating the number of student selections per project).
3. Simplified reports for administrators, such as the number of unmatched students and project capacity.

All of the above graphics would be implemented using basic graphing software compatible with the existing React framework, without interfering with the existing matching algorithm.

8.3 NOTIFICATION AND ANNOUNCEMENT CENTER

Notifications Inside the Application For now, users need to refresh the pages in order to receive updates on the state of projects. In order to avoid any misunderstandings, a notifications center needs to be implemented inside the application.

Some of the features of the improved system will include:

- The notifications section within each of the portals where system announcements will be published (phase change, project approval, deadlines, etc.).
- The simple inbox of each user displaying their read and unread notifications.
- The notification icons informing users about new notifications.

Notifications will be inside the application, thus not requiring any emails.

8.4 IMPROVEMENTS TO TEACHER AND ADMIN PORTALS

However, the current Teacher and Admin portals represent the basic layout and the main page of those portals, although there are some unfinished features. Thus, completing and perfecting current processes rather than implementing new systems is what needs to be done at this stage.

Specifically, improvements that should be introduced for the Teacher portal include:

- Improving functionality related to project management (such as creation, editing, and archiving of projects).
- Improving the applicant view to allow the supervisor to see such data as the GPA, department, and preference.
- Implementing a feature that would allow writing notes to the applicants' profile (e.g., for comments on particular students/projects).

As for the Admin portal, the following improvements should be made:

- Integrating status indicators to reflect whether the application was approved/rejected/pending.
- Drafting an additional feature for manually assigning students that did not get assigned automatically.
- Enhancing deadline management to simplify working with phase start/end dates.

These changes are expected to be beneficial for the system.

8.5 USABILITY IMPROVEMENTS AND USER FEEDBACK

Although the existing interfaces function well, not all areas of usability testing have been done by users. Future work would entail small enhancements based on feedback from users.

Actions to be taken would include:

- Conducting user feedback sessions with a small number of students, supervisors, and administrators to identify unclear processes and omitted details.
- Enhancing the design of forms, labels, and validation errors in order to prevent inaccurate entries and to clarify every process.
- Including instructions (e.g., tool tips) on critical pages, such as the preference entry page and matching output page.

The planned improvements will be concentrated on usability issues that simplify learning and operation of the system, especially for new users.

8.6 BASIC TESTING AND DEPLOYMENT PREPARATION

Up until now, most testing has been done manually during the course of development. In order to give us more confidence about the stability of the system, some testing and preparation for deployment will be done.

What this will consist of:

- A few unit tests to be performed on the backend components that matter, including the matching process as well as key APIs.
- Conducting small load tests under conditions when the system is expected to be stressed, such as near deadline when there is high traffic from students submitting/altering selections.
- Prepare the system for deployment on the test server by getting ready everything necessary for a production version.

None of the above will require any sophisticated infrastructure.

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